

Durnesswick Laboratories Inc

NM-12 Oral Vehicle: Precipitation-on-Dilution Follow-up

Final Report

Work package DL_2020_0050

Client: Nimbravelle Medicines Inc

Date: 09/28/2020

Prepared by: Dr J. Carver, Formulation Development Lead

1. Summary

Nimbravelle needed a 10 mg/mL oral vehicle for NM-12 for repeat-dose rodent studies. The cosolvent vehicle selected earlier (PEG400/water 30:70) gave a clear solution at 10 mg/mL but precipitated within minutes when diluted into simulated gastric fluid. We recommend, provisionally, a dispersed suspension in 0.5% methylcellulose, which was easily redispersed after 24 hours. Chemical stability and dose homogeneity over the dosing period must be confirmed before the first study.

2. Materials

- NM-12 drug substance, lot NM12-DS-04 (client CoA), micronised
- Excipients: methylcellulose (400 cP grade), hydroxypropyl methylcellulose, polysorbate 80, PEG400, hydroxypropyl- β -cyclodextrin, all compendial grade
- Simulated gastric fluid without pepsin (SGF), pH 1.2, prepared fresh

3. Vehicle ladder

Eight vehicles were screened. Equilibrium solubility was measured by shake-flask; appearance and 24-hour physical stability were assessed on separately prepared 10 mg/mL formulations. This is an exploratory screen, not a locked process.

Table 1 Vehicle ladder results

Vehicle	Equilibrium solubility (mg/mL)	Appearance at 10 mg/mL loading	24 h physical stability (10 mg/mL)
0.5% methylcellulose	0.31	Fine suspension, easily redispersed	Redisperses on shaking
0.5% HPMC / 0.1% Tween 80	1.2	Suspension, settles <1 h	Crystal growth at 24 h
20% HP- β -CD in water	1.7	Fine suspension	Crystal growth at 24 h
PEG400/water 30:70	14	Clear solution	Stable
10% DMSO / 90% corn oil	2.1	Suspension, settles <1 h	Crystal growth at 24 h
Labrasol/PEG400 1:1	8.8	Fine suspension, easily redispersed	Redisperses on shaking
5% NMP / 15% Solutol HS15 / 80% water	1.2	Fine suspension, easily redispersed	Caking at 24 h
pH 4 citrate buffer	3.2	Fine suspension	Caking at 24 h

Solubility by shake-flask, 24 h, 25 °C, HPLC-UV; appearance and stability assessed on separately prepared 10 mg/mL formulations

4. Dilution challenge

Each 10 mg/mL formulation was diluted 1:10 into SGF at 37 °C with gentle stirring and observed for 30 minutes against a black and a white background. Vehicle-only dilutions were run alongside as controls.

Table 2 Dilution observations

Vehicle (10 mg/mL)	Observation after dilution into SGF
PEG400/water 30:70	Turbid within 2 min; crystalline precipitate by 10 min
Labrasol/PEG400 1:1	Fine haze at 5 min, settling by 30 min
0.5% methylcellulose	Particles dispersed; no visible change in particle appearance
20% HP- β -CD in water	Suspended material unchanged

The cosolvent vehicles precipitated on dilution, so the earlier solution approach is rejected.

5. Recommended vehicle (provisional)

Table 3 Recipe card, 10 mg/mL

Component	Per 100 mL
NM-12 (micronised)	1.00 g
Methylcellulose 400 cP	0.50 g
Purified water	to 100 mL

- Disperse methylcellulose in one third of the water at 80 °C, then add the remaining water cold and stir until clear
- Wet the NM-12 powder with a small volume of vehicle to a paste, then add the rest in portions with stirring
- Stir continuously during dosing; dose volume 10 mL/kg
- Before the first dose, sample top, middle and bottom and confirm homogeneity by HPLC

6. Limitations and next steps

- Chemical stability of NM-12 in the vehicle beyond the day of preparation was not tested; until it is, prepare fresh daily
- Dose homogeneity and redispersibility were assessed visually only
- Particle size was not controlled; if exposure is variable in the first study, a milled suspension should be evaluated

Name	Role	Signature	Date
Julia Carver	Formulation Development Lead	<u>J Carver</u>	09/28/2020
Miguel Serrano	Senior Formulation Scientist	<u>M Serrano</u>	09/28/2020